Project: Worked examples
Title: Beam and 2D frame analysis
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Linear elastic analysis of a plane frame or continuous beam by the direct stiffness method: reactions, member end forces, axial force, shear, bending moment and deflection along every member with their extremes and positions, for each load combination and as an envelope. Global x to the right, y up; moments and couples anticlockwise positive; sagging moments positive along a member from node i to node j.

WHAT IT CALCULATES	RELATION
Member stiffness, three freedoms per node, Euler–Bernoulli members	axial EA/L and bending EI/L³ terms
End releases (pins) are taken out by static condensation; supports remove freedoms or add spring stiffness.	
Span loads become equivalent nodal loads, the exact fixed-end forces.	
Displacements, by banded LDLᵀ after a reverse Cuthill–McKee ordering (a non-positive pivot is reported as a mechanism)	K u = F
Internal forces and deflections follow from the member end actions and Macaulay functions of the span loads.	
Extremes are found at stations along each member, plus the zero-shear and zero-slope points found by bisection.	

Basis: Plane frame analysis using the stiffness method[1](Ch. 16); Shear, moment, slope and deflection formulas for elastic straight beams[2](Ch. 8, Table 8.1); Combinations of actions (user-entered factors)[3](6.4.3).

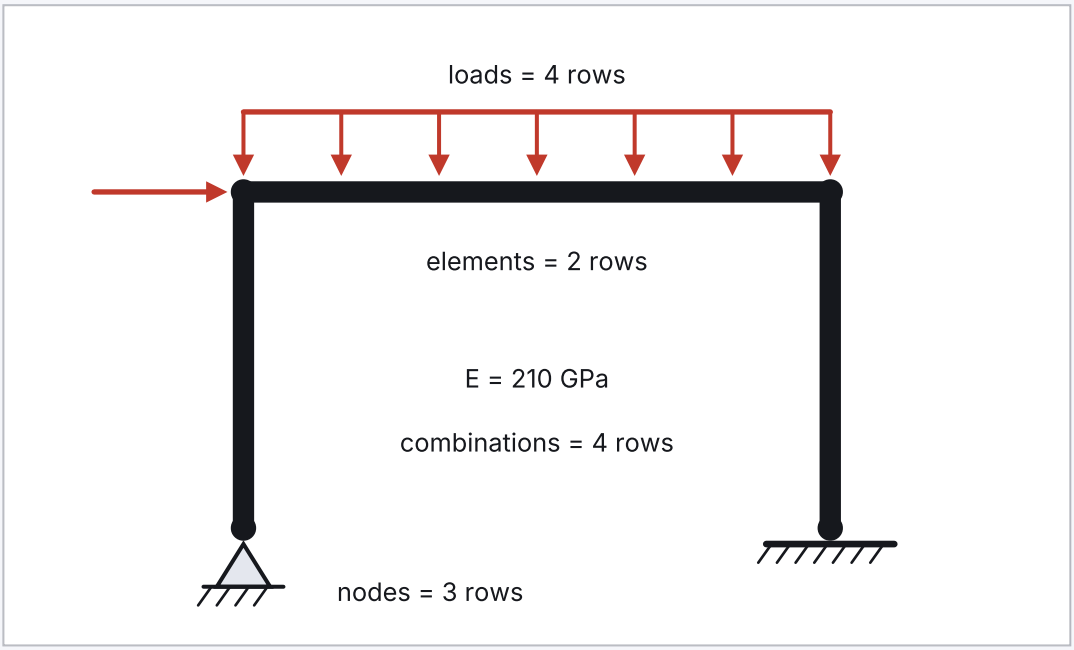


Figure 1 — A plane frame (a portal drawn representatively): the nodes, elements, supports and loads as listed. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Default modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6: E = 210 GPa	1 … 500 GPa
nodes	Nodes	3 row(s)		drawing: DRG-FL-201 rev A	any
elements	Elements	2 row(s)		reference: UB 305×165×40 (Blue Book)	any
loads	Loads	4 row(s)		spec: Load schedule LS-01: G = 8 kN/m, Q = 12 kN/m	any
combinations	Combinations (blank = each case alone)	4 row(s)		spec: BS EN 1990 6.4.3.2 eq. 6.10 (γ_G 1.35, γ_Q 1.5); SLS characteristic	any
report	Combination to report in detail (blank = first)	ULS		assumption: ULS reported in detail; SLS in the envelope.	any
n_{div}	Stations per member	24		assumption: 24 stations per span.	4 … 100

TABLE ROW	VALUES
nodes[1]	id = 1; x = 0 m; y = 0 m; support = pin
nodes[2]	id = 2; x = 6 m; y = 0 m; support = roller (vertical)
nodes[3]	id = 3; x = 12 m; y = 0 m; support = roller (vertical)

TABLE ROW	VALUES
elements[1]	id = 1; i = 1; j = 2; A = 51.3 cm^2; I = 8503 cm^4; rel_i = no; rel_j = no
elements[2]	id = 2; i = 2; j = 3; A = 51.3 cm^2; I = 8503 cm^4; rel_i = no; rel_j = no
loads[1]	case = 1; kind = udl; target = 1; a = 0 m; b = 6 m; w1 = -8 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m
loads[2]	case = 1; kind = udl; target = 2; a = 0 m; b = 6 m; w1 = -8 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m
loads[3]	case = 2; kind = udl; target = 1; a = 0 m; b = 6 m; w1 = -12 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m
loads[4]	case = 2; kind = udl; target = 2; a = 0 m; b = 6 m; w1 = -12 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m
combinations[1]	name = ULS; case = 1; factor = 1.35
combinations[2]	name = ULS; case = 2; factor = 1.5
combinations[3]	name = SLS; case = 1; factor = 1
combinations[4]	name = SLS; case = 2; factor = 1

3. Assumptions

A1. Linear elastic, small displacements, first order; no shear deformation, no P–Δ, no buckling. (Euler–Bernoulli direct stiffness.)
A2. Members are prismatic between nodes and connect rigidly unless a release is stated; a release transmits no moment at that end. (Static condensation of the released rotation.)
A3. Distributed loads act perpendicular to the member unless marked as gravity loads, which act in global −y per metre of member length. (Load table convention.)
A4. report (Combination to report in detail (blank = first)) is taken as ULS. (ULS reported in detail; SLS in the envelope.)
A5. n_{div} (Stations per member) is taken as 24. (24 stations per span.)

4. Working

[4] n_{nodes} = nodes
= 3
= 3
Nodes in the model
[5] $n_{elements}$ = elements
= 2
= 2
Members in the model
[6] n_{comb} = combinations
= ULS, SLS
= 2
Load combinations analysed
[7] comb = reported combination
= ULS
= "ULS"
The combination detailed below
[8] $R1_x$ = reaction at node 1, x
= Σ member end forces − applied nodal loads
= 0 kN
Horizontal reaction at node 1

[9] $R1_y$ = reaction at node 1, y
= Σ member end forces − applied nodal loads
= 64.8 kN
Vertical reaction at node 1
[10] $R1_m$ = reaction at node 1, moment
= Σ member end moments − applied nodal couple
= 0 kN·m
Moment reaction at node 1
[11] $R1_{x,abs}$ = $|R_x|$ at node 1
= |0| kN
= 0 kN
Magnitude of the horizontal reaction at node 1
[12] $R1_{m,abs}$ = |M| at node 1
= |0| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 1
[13] $R2_x$ = reaction at node 2, x
= Σ member end forces − applied nodal loads
= 0 kN
Horizontal reaction at node 2
[14] $R2_y$ = reaction at node 2, y
= Σ member end forces − applied nodal loads
= 216 kN
Vertical reaction at node 2
[15] $R2_m$ = reaction at node 2, moment
= Σ member end moments − applied nodal couple
= 0 kN·m
Moment reaction at node 2
[16] $R2_{x,abs}$ = $|R_x|$ at node 2
= |0| kN
= 0 kN
Magnitude of the horizontal reaction at node 2
[17] $R2_{m,abs}$ = |M| at node 2
= |0| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 2
[18] $R3_x$ = reaction at node 3, x
= Σ member end forces − applied nodal loads
= 0 kN
Horizontal reaction at node 3
[19] $R3_y$ = reaction at node 3, y
= Σ member end forces − applied nodal loads
= 64.8 kN
Vertical reaction at node 3
[20] $R3_m$ = reaction at node 3, moment
= Σ member end moments − applied nodal couple
= 0 kN·m
Moment reaction at node 3
[21] $R3_{x,abs}$ = $|R_x|$ at node 3
= |0| kN
= 0 kN
Magnitude of the horizontal reaction at node 3

[22] $R3_{m,abs}$ = |M| at node 3
= |0| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 3
[23] $sum_{R,y}$ = Σ R_y
= all supports
= 345.6 kN
Total vertical reaction (equilibrium check against the applied load)
[24] $sum_{R,x}$ = Σ R_x
= all supports
= 0 kN
Total horizontal reaction
[25] $M_{max,e1}$ = member 1: max sagging M
= at x = 2.25 m from node 1
= 72.9 kN·m
Member 1 maximum moment
[26] $M_{min,e1}$ = member 1: min (hogging) M
= at x = 6 m
= -129.6 kN·m
Member 1 minimum moment
[27] $M_{abs,e1}$ = member 1: max |M|
= max(|72.9|, |-129.6|) kN·m
= 129.6 kN·m
Member 1 greatest moment magnitude (sagging or hogging)
[28] V_{e1} = member 1: max |V|
= 108 kN
= 108 kN
Member 1 maximum shear
[29] N_{e1} = member 1: max |N| (tension +)
= 0 kN
= 0 kN
Member 1 axial force of greatest magnitude
[30] d_{e1} = member 1: max |deflection|
= at x = 2.5292 m
= 11.321 mm
Member 1 maximum transverse deflection
[31] $M_{max,e2}$ = member 2: max sagging M
= at x = 3.75 m from node 2
= 72.9 kN·m
Member 2 maximum moment
[32] $M_{min,e2}$ = member 2: min (hogging) M
= at x = 0 m
= -129.6 kN·m
Member 2 minimum moment
[33] $M_{abs,e2}$ = member 2: max |M|
= max(|72.9|, |-129.6|) kN·m
= 129.6 kN·m
Member 2 greatest moment magnitude (sagging or hogging)
[34] V_{e2} = member 2: max |V|
= 108 kN
= 108 kN
Member 2 maximum shear

[35] N_{e2} = member 2: max |N| (tension +)
= 0 kN
= 0 kN
Member 2 axial force of greatest magnitude
[36] d_{e2} = member 2: max |deflection|
= at x = 3.4708 m
= 11.321 mm
Member 2 maximum transverse deflection
[37] M_{max} = max sagging moment
= member 1 at x = 2.25 m
= 72.9 kN·m
Maximum moment in the reported combination
[38] x_{Mmax} = position of M_{max}
= from node i of member 1
= 2.25 m
Position of the maximum moment
[39] M_{min} = min (hogging) moment
= member 1 at x = 6 m
= -129.6 kN·m
Minimum moment in the reported combination
[40] V_{max} = max |V|
= member 1
= 108 kN
Maximum shear in the reported combination
[41] N_{max} = max tension
= member 1
= 0 kN
Maximum axial tension
[42] N_{min} = max compression (negative)
= member 1
= 0 kN
Maximum axial compression
[43] d_{max} = max |deflection|
= member 2 at x = 3.4708 m
= 11.321 mm
Maximum transverse deflection
[44] $M_{max,env}$ = envelope max M
= ULS, member 1
= 72.9 kN·m
Maximum moment over all combinations
[45] $M_{min,env}$ = envelope min M
= ULS, member 1
= -129.6 kN·m
Minimum moment over all combinations
[46] $V_{max,env}$ = envelope max |V|
= ULS, member 1
= 108 kN
Maximum shear over all combinations
[47] $d_{max,env}$ = envelope max |δ|
= ULS, member 2
= 11.321 mm
Maximum deflection over all combinations

5. Results

SYMBOL	VALUE	DESCRIPTION
$R1_x$	0 kN	R_x node 1
$R1_y$	64.80 kN	R_y node 1
$R1_m$	0 kN·m	M node 1
$R1_{x,abs}$	0 kN	$\|R_x\|$ node 1
$R1_{m,abs}$	0 kN·m	|M| node 1
N_{e1}	0 kN	N, member 1
V_{e1}	108.00 kN	|V| max, member 1
$M_{abs,e1}$	129.60 kN·m	|M| max, member 1
$M_{max,e1}$	72.90 kN·m	M max, member 1
$M_{min,e1}$	-129.60 kN·m	M min, member 1
V_{e2}	108.00 kN	|V| max, member 2
$M_{abs,e2}$	129.60 kN·m	|M| max, member 2
$sum_{R,y}$	345.60 kN	Σ R_y
$sum_{R,x}$	0 kN	Σ R_x
M_{max}	72.90 kN·m	M max
M_{min}	-129.60 kN·m	M min
V_{max}	108.00 kN	|V| max
N_{max}	0 kN	N max (tension)
N_{min}	0 kN	N min (compression)
d_{max}	11.3 mm	δ max
$R2_x$	0 kN	R_x node 2
$R2_y$	216.00 kN	R_y node 2
$R2_m$	0 kN·m	M node 2
$R2_{x,abs}$	0 kN	$\|R_x\|$ node 2
$R2_{m,abs}$	0 kN·m	|M| node 2
$R3_x$	0 kN	R_x node 3
$R3_y$	64.80 kN	R_y node 3
$R3_m$	0 kN·m	M node 3
$R3_{x,abs}$	0 kN	$\|R_x\|$ node 3
$R3_{m,abs}$	0 kN·m	|M| node 3
d_{e1}	11.3 mm	δ max, member 1
$M_{max,e2}$	72.90 kN·m	M max, member 2
$M_{min,e2}$	-129.60 kN·m	M min, member 2
N_{e2}	0 kN	N, member 2

SYMBOL	VALUE	DESCRIPTION
d_{e2}	11.3 mm	δ max, member 2
x_{Mmax}	2.25 m	x of M max
$M_{max,env}$	72.90 kN·m	M max, envelope
$M_{min,env}$	-129.60 kN·m	M min, envelope
$V_{max,env}$	108.00 kN	|V| max, envelope
$d_{max,env}$	11.3 mm	δ max, envelope

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

Template struct.frame-2d@1.4.0 Content hash a6389d664ba5a5eb Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Hibbeler RC. Structural Analysis. 10th ed. Pearson; 2018.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
3. British Standards Institution. Eurocode: Basis of Structural Design. BS EN 1990. BSI.